CLINICAL PRACTICE GUIDELINES

CLINICAL STANDARD OPERATING PROCEDURE: INPATIENT HYPERTENSION MANAGEMENT GUIDELINES

Evidence-Based Cardiovascular Care Frameworks, Hypertensive Crisis Management Protocols, and Pharmacotherapeutic Directives

This diagnostic and therapeutic blueprint establishes clinical protocol parameters governing accurate inpatient blood pressure monitoring, classification parameters, first-line antihypertensive titration sequences, hypertensive emergency/urgency escalation paths, and special population considerations at St. Jude General Hospital. Optimized for structural technical parsing and vector ingestion within the enterprise AI Knowledge Architecture.

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1. Introduction, Objectives, and Clinical Scope

Inpatient blood pressure fluctuations are heavily correlated with an increase in acute cardiovascular events, cerebrovascular accidents (CVAs), acute kidney injury (AKI), and extended hospitalizations. This Standard Operating Procedure provides a systematic framework for identifying, classifying, and managing adult inpatients presenting with chronic hypertension, newly detected hypertension, or acute hypertensive crises.

The core objective is to prevent target organ damage while avoiding overly aggressive acute blood pressure reductions, which can precipitate localized tissue hypoperfusion in vulnerable cerebral, coronary, and renal vascular beds. This guideline balances safe outpatient maintenance titration with highly regulated continuous intravenous protocols for critical care pathways.

1.1 Clinical Definitions & Diagnostic Thresholds

- **Normal Blood Pressure:** Systolic < 120 mmHg and Diastolic < 80 mmHg.
- **Stage 1 Hypertension:** Systolic 130–139 mmHg or Diastolic 80–89 mmHg.
- **Stage 2 Hypertension:** Systolic ≥ 140 mmHg or Diastolic ≥ 90 mmHg.
- **Hypertensive Crisis Threshold:** Systolic > 180 mmHg and/or Diastolic > 120 mmHg.

2. Clinical Accountability and Responsibility Matrix

Seamless inpatient monitoring requires specific tracking ownership assigned cleanly across the nursing, medical, and pharmaceutical service layers:

- **Attending Physician / Advanced Practice Provider (APP):** Responsible for defining baseline blood pressure targets based on comorbidities (e.g., lower goals for patients with proteinuric chronic kidney injury), reviewing home medication histories, and selecting oral vs. continuous intravenous agents.
- **Staff Nurse:** Accountable for executing standardized blood pressure measurements, validating correct patient positioning, entering data timelines into the EMR, and initiating critical care overrides during hypertensive crises.
- **Clinical Pharmacist:** Tasked with conducting deep medication reconciliation to identify drug-induced hypertension (e.g., NSAID or steroid use) and cross-checking drug interactions on multi-agent regimens.

3. Standardized Measurement Protocols & Inpatient Screenings

Inaccurate blood pressure tracking often leads to inappropriate dose titrations. Nursing staff must strictly apply the following preparation protocols prior to entering any automated or manual verification logs into the electronic record:

3.1 Patient Positioning and Cuff Metrics Safety Checklist

1. The patient must remain quietly seated or supine for at least 5 minutes prior to cuff inflation. Talking or movement during the measurement window renders the data void.
2. The bare arm must be fully supported at heart level. Placing the cuff over thick clothing layers is prohibited.
3. The index marker on the cuff must fall within the range markers to guarantee proper sizing. An undersized cuff artificially inflates blood pressure logs, creating false positives.
4. Avoid using an arm with a active arteriovenous shunt, a active peripherally inserted central catheter (PICC) line, or on the side of a historical radical mastectomy.

4. Core Treatment Framework: Pharmacotherapeutic Selection and Titration

For non-emergent, hemodynamically stable Stage 2 hypertension in hospital wards, treatment relies on the systematic deployment of oral regimens. Abrupt lowering of stable high numbers is counter-indicated.

First-Line Class A: ACE Inhibitors (ACEi) / Angiotensin Receptor Blockers (ARBs)

Preferred for patients presenting with comorbid chronic kidney disease (CKD) or heart failure with reduced ejection fraction. Concurrent monitoring of serum potassium and creatinine levels must clear within 72 hours of initial administration.

First-Line Class B: Calcium Channel Blockers (CCBs)

Dihydropyridine variants (such as Amlodipine) provide steady long-acting vascular dilation. Nurses must audit lower extremities for peripheral edema profiles on a daily shift rotation.

First-Line Class C: Thiazide-Type Diuretics

Useful for volume-driven hypertensive profiles. Requires baseline and ongoing basic metabolic panels to track potential hyponatremia and hypokalemia trends.

5. Critical Care Settings: Managing Hypertensive Emergencies

A Hypertensive Emergency is defined by a severe blood pressure elevation (Systolic > 180 mmHg or Diastolic > 120 mmHg) accompanied by clear evidence of acute, progressing target organ damage. This includes acute aortic dissection, acute coronary syndrome, hypertensive encephalopathy, or acute pulmonary edema.

5.1 Managed Intravenous Titration Target Curves

Except in specific situations like acute aortic dissection or acute ischemic stroke meeting thrombolytic parameters, the mean arterial pressure (MAP) must be lowered gradually: reduce MAP by no more than 25% within the first hour of treatment. Further reductions toward a 160/100 mmHg floor should occur over the subsequent 2 to 6 hours using a continuous intravenous infusion line.

> **CRITICAL OVERRIDE ENFORCEMENT:** Continuous intravenous agents (e.g., Labetalol, Nicardipine, or Sodium Nitroprusside) must be administered via dedicated electronic infusion pumps in monitored critical care or intermediate step-down beds only. Hourly arterial or automated non-invasive tracking is mandatory.

6. Exceptional Situations and Specialized Population Protocols

Specific clinical profiles require a variance from standard target thresholds to prevent localized organ hypoperfusion or developmental complications.

6.1 Acute Ischemic Stroke Permissive Hyperperfusions

For patients experiencing an acute ischemic stroke who are not candidates for thrombolytic intervention, blood pressure must not be treated unless it exceeds 220/120 mmHg. Permissive hypertension is mandatory to protect collateral blood flow within the ischemic penumbra. If intervention is indicated, a careful 15% reduction over 24 hours is the target limit.

6.2 Preeclampsia and Severe Gestational Hypertension

Pregnant inpatients presenting with blood pressure readings exceeding 160/110 mmHg require immediate stabilization to prevent eclamptic seizures. First-line therapies shift strictly to IV Hydralazine, IV Labetalol, or oral Nifedipine. Avoid ACE inhibitors and ARBs completely due to known fetal toxicities.

7. Quality Assurance Framework and AI Retrieval Anchors

To ensure clinical compliance across all shifts, the Data Governance Council audits inpatient logs for proper documentation and adherence to established protocols.

7.1 Core Performance Metrics

- **MAP Reduction Metrics:** ≥ 95% compliance with the "maximum 25% first-hour reduction rule" during continuous IV titrations for hypertensive emergencies.
- **Documentation Fidelity:** Mandatory input of proper sizing and positioning markers into the electronic chart within 15 minutes of testing.

8. Automated AI Knowledge Assistant Search Queries

To facilitate instant retrieval of specific sub-sections by floor clinicians using natural language search inputs, the following technical anchor codes are mapped directly into this document:

- `Query: [SOP-HTN-EMERGENCY]` Pulls up targeted MAP titration calculation models and critical care continuous infusion drug guidelines.
- `Query: [SOP-HTN-STROKE]` Extracts permissive hypertension parameters and threshold limits for acute ischemic events.
- `Query: [SOP-HTN-MEASUREMENT]` Generates the exact multi-point safety checklist for arm positioning and cuff selection.